Digital Image Processing: Digital Imaging Fundamentals

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Contents

- This lecture will cover:
  - The human visual system
  - Light and the electromagnetic spectrum
  - Image representation
  - Image sensing and acquisition
  - Sampling, quantisation and resolution

Human Visual System

- The best vision model we have!
- Knowledge of how images form in the eye can help us with processing digital images
- We will take just a whirlwind tour of the human visual system

Structure Of The Human Eye

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- The lens focuses light from objects

onto the retina

- The retina is covered with light receptors called *cones* (6-7 million) and *rods* (75-150 million)
- Cones are concentrated around the fovea and are very sensitive to colour
- Rods are more spread out and are sensitive to low levels of illumination

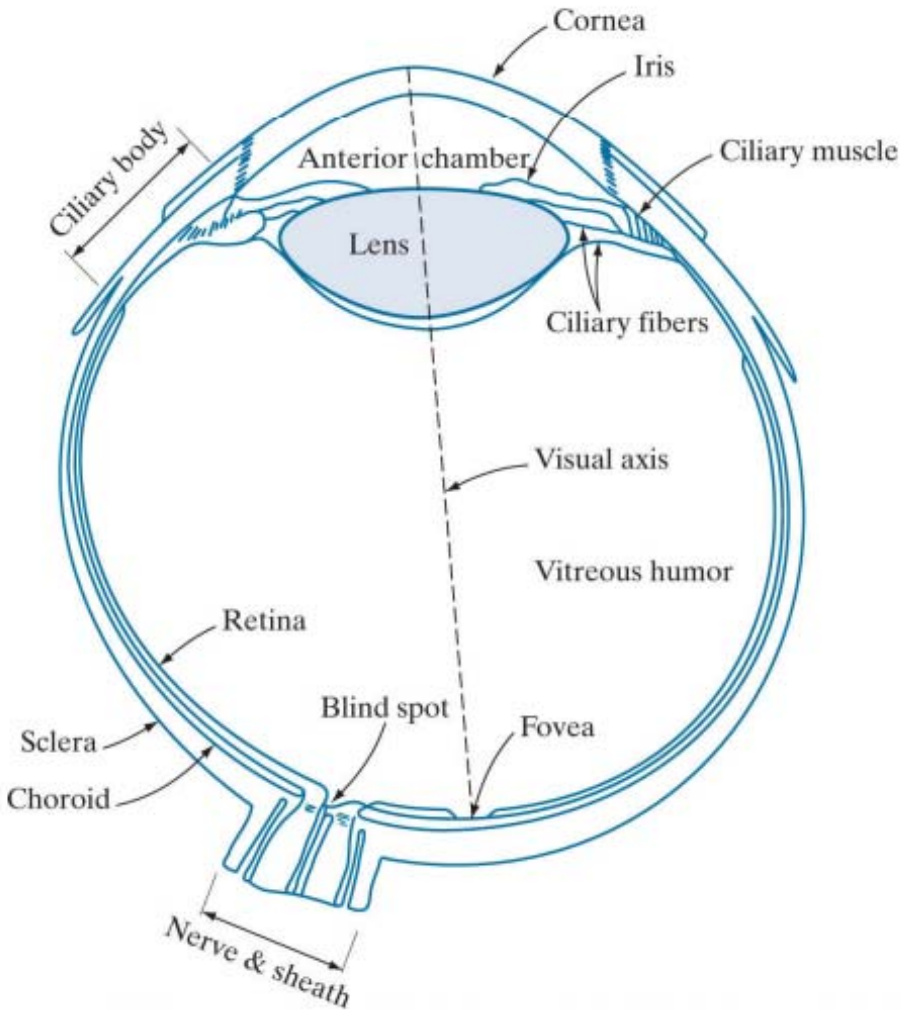
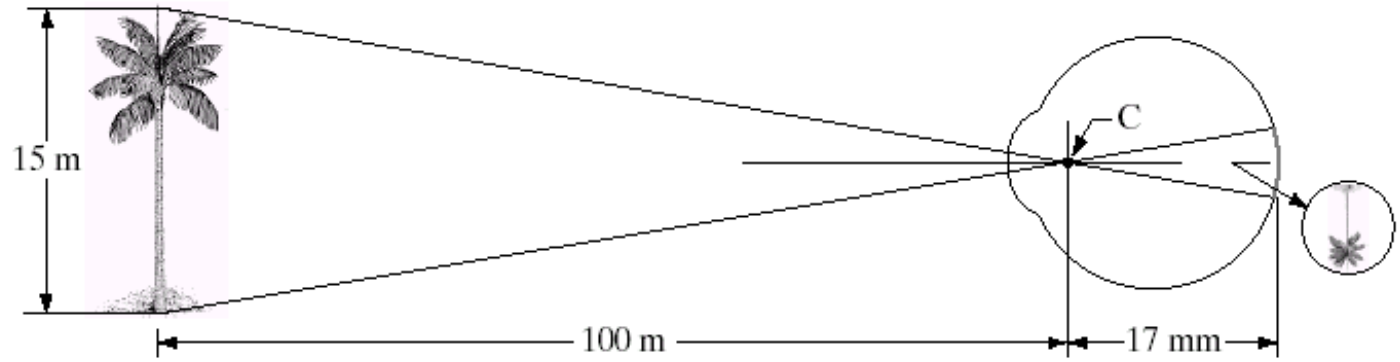


Image Formation In The Eye

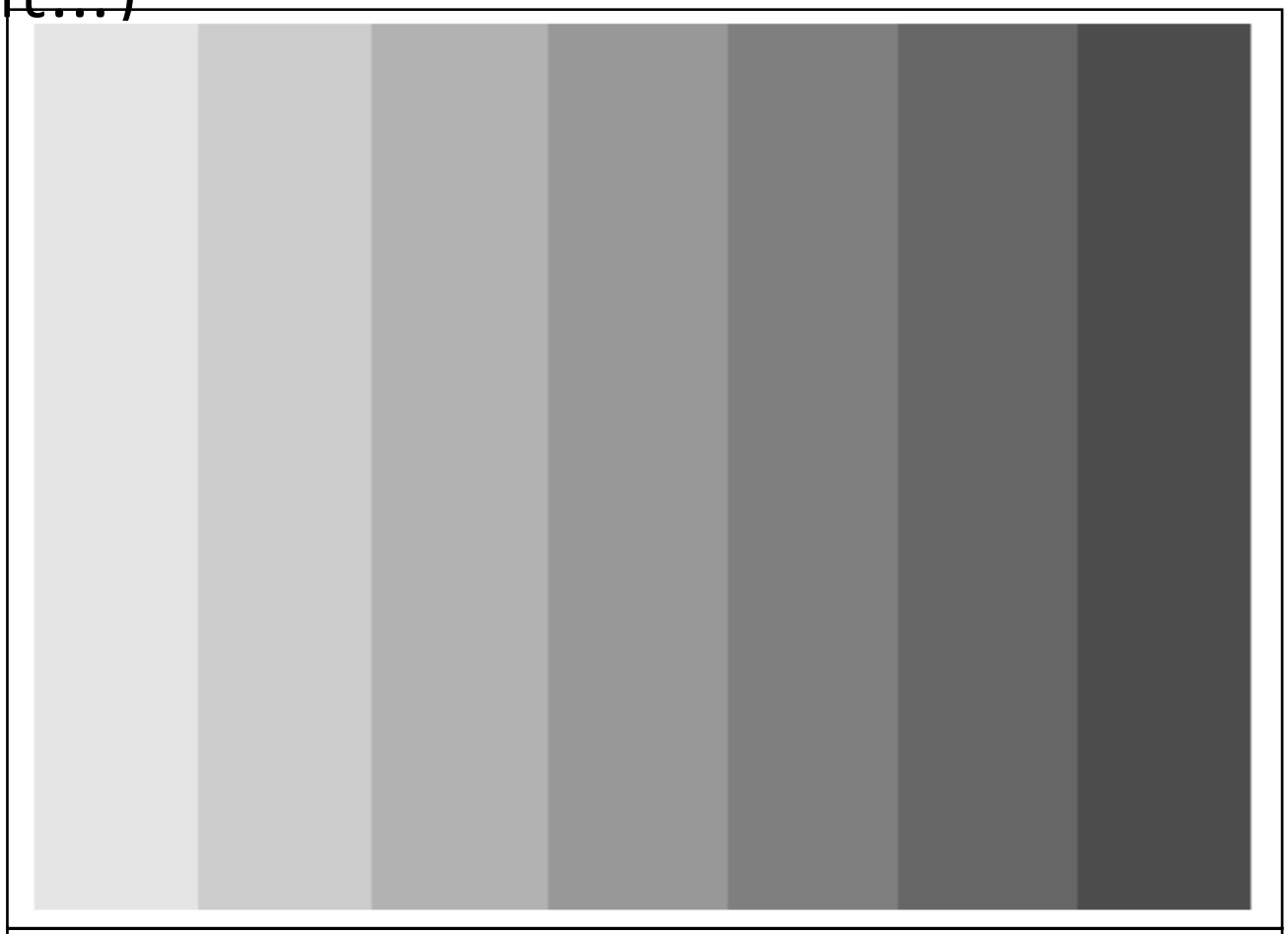
- Muscles within the eye can be used to change the shape of the lens allowing us focus on objects that are near or far away
- An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain



Brightness Adaptation & Discrimination

- The human visual system can perceive approximately 10^{10} different light intensity levels
- However, at any one time we can only discriminate between a much smaller number – *brightness adaptation*
- Similarly, the *perceived intensity* of a region is related to the light intensities of the regions surrounding it

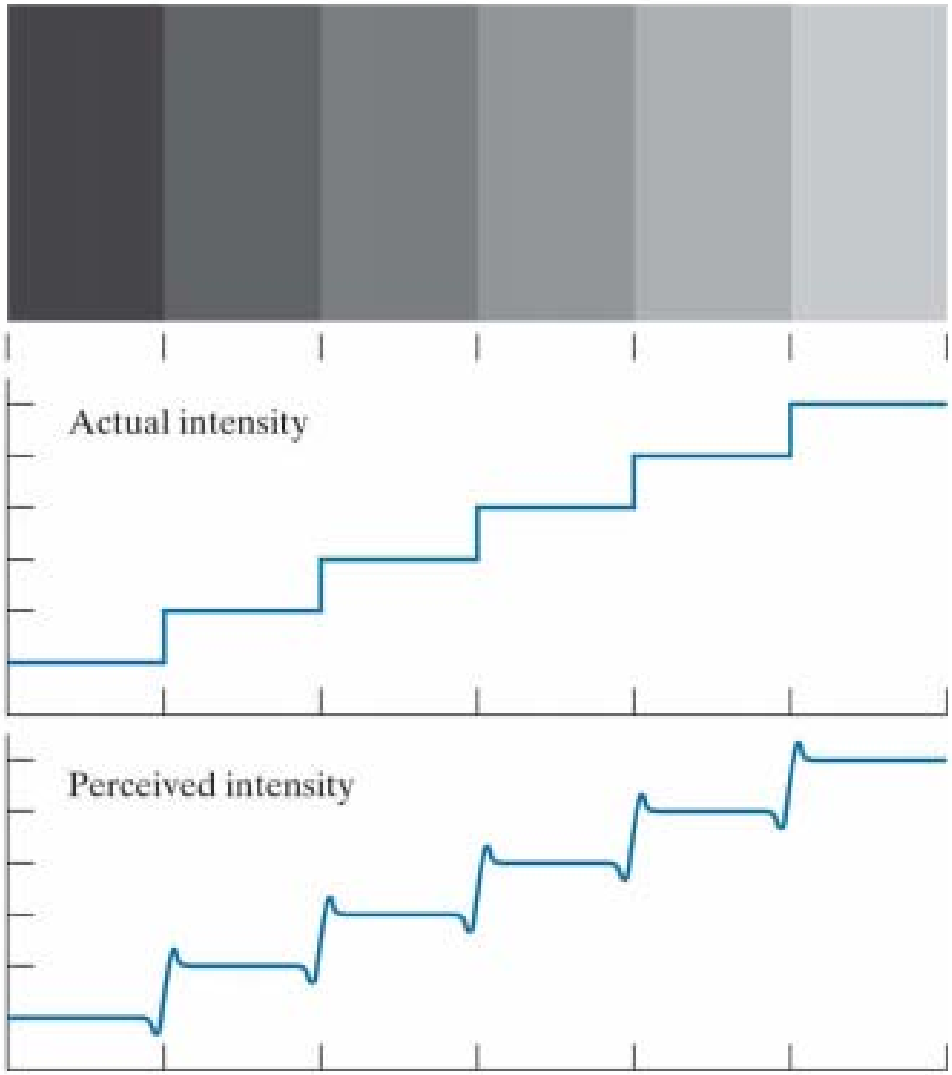
Brightness Adaptation & Discrimination (cont…)



An example of Mach bands

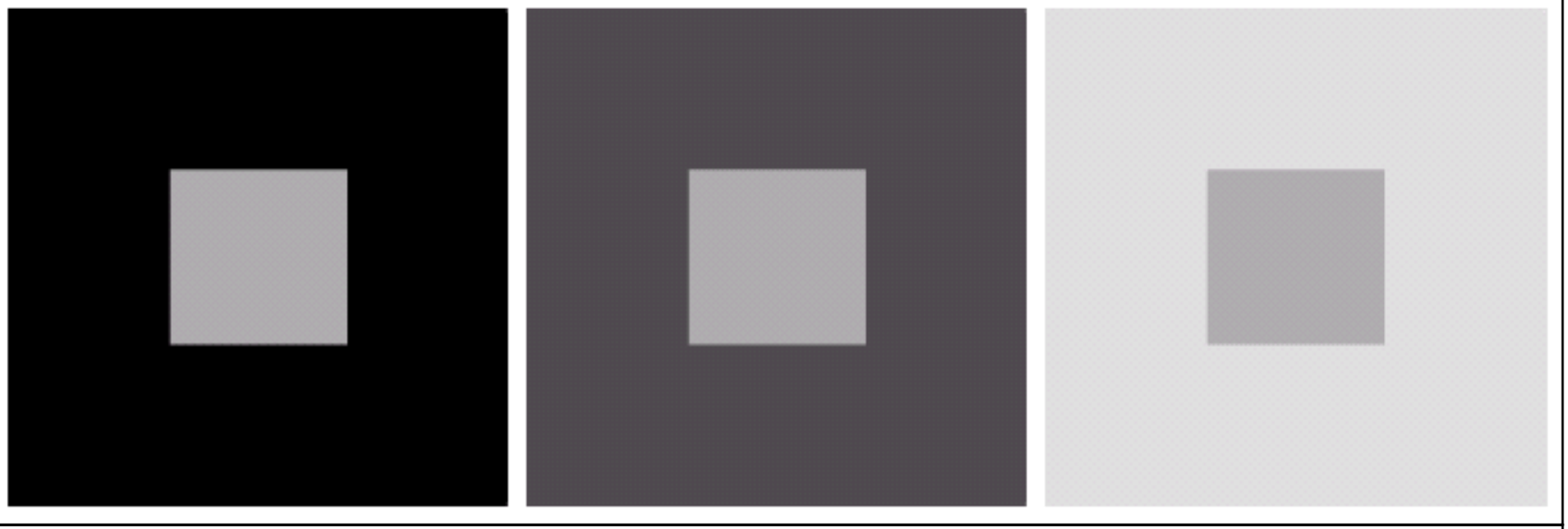
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Brightness Adaptation & Discrimination (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Brightness Adaptation & Discrimination (cont…)

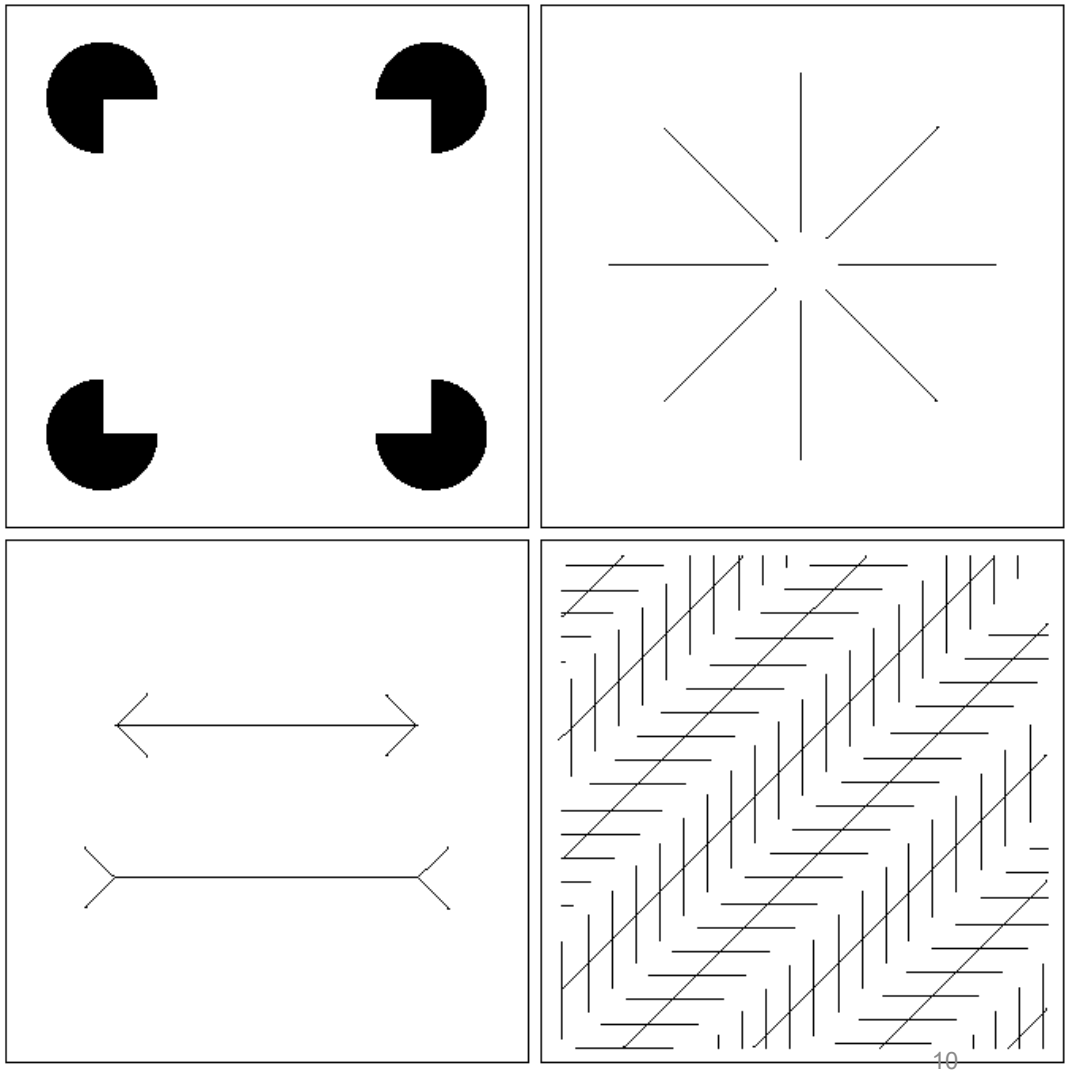


An example of *simultaneous contrast*

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Optical Illusions

•Our visual systems play lots of interesting tricks on us



Light And The Electromagnetic Spectrum

- Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye
- The electromagnetic spectrum is split up according to the wavelengths of different forms of energy

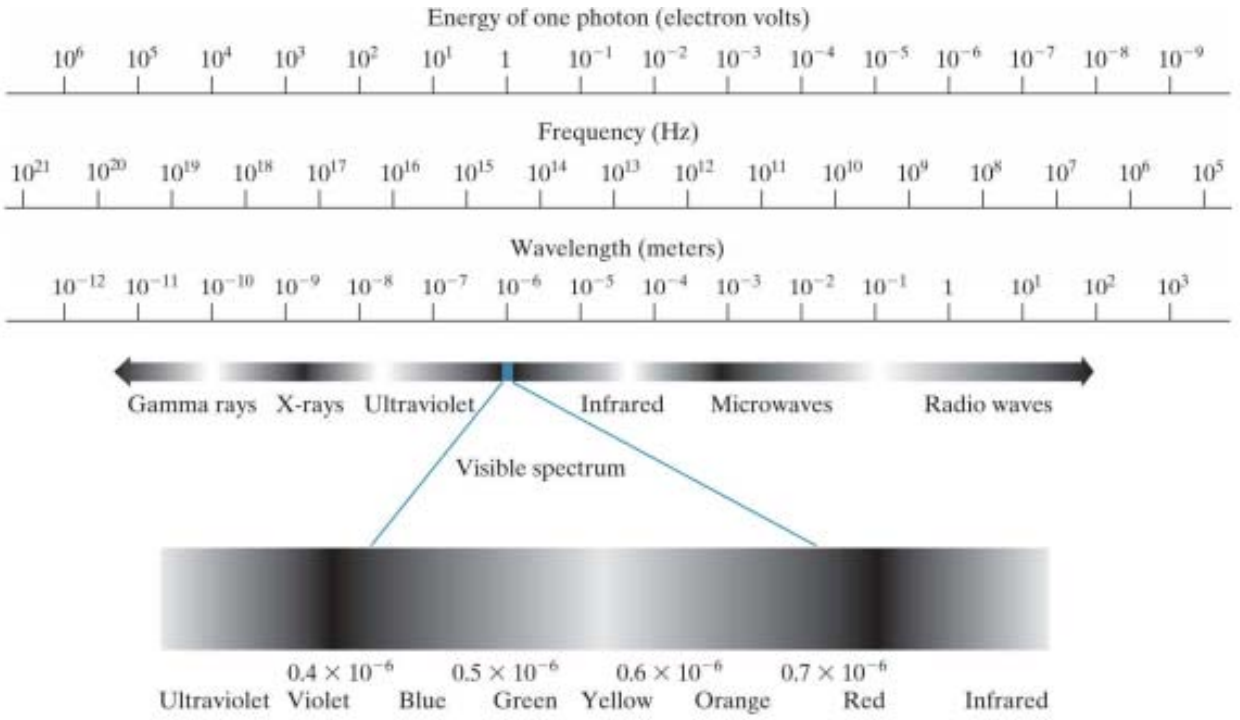
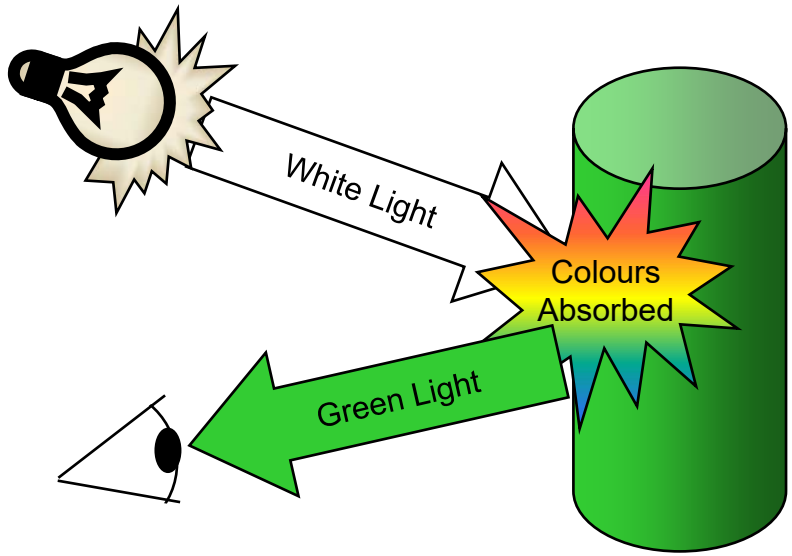


FIGURE 2.10

The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.

Reflected Light

- The colours that we perceive are determined by the nature of the light reflected from an object
- For example, if white light is shone onto a green object most wavelengths are absorbed, while green light is reflected from the object

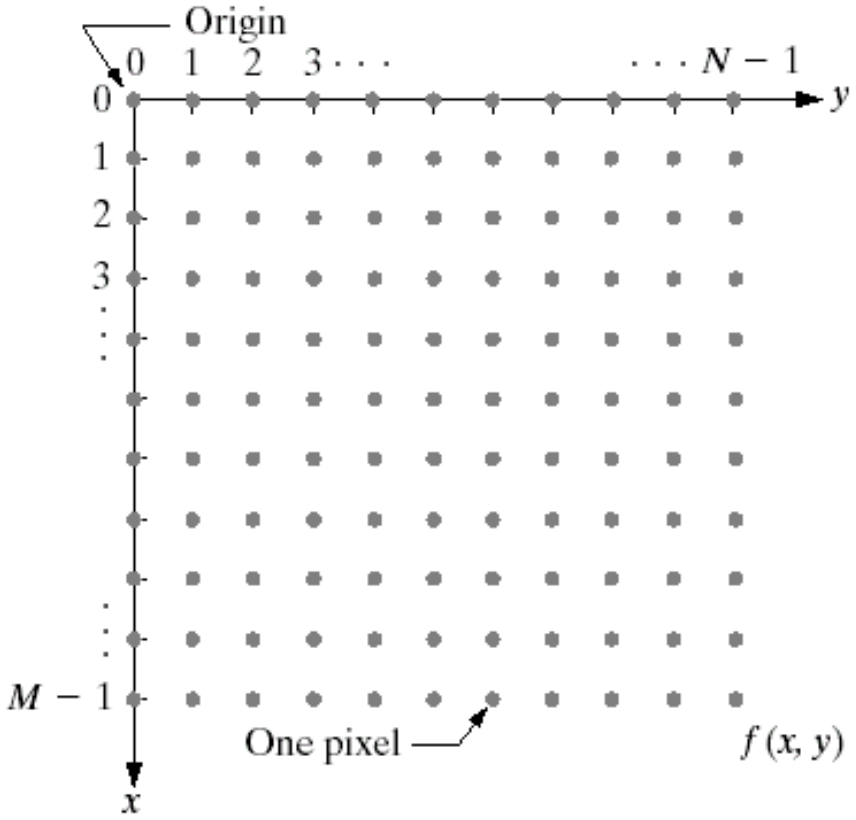


Sampling, Quantisation And Resolution

- In the following slides we will consider what is involved in capturing a digital image of a real-world scene
  - Image sensing and representation
  - Sampling and quantisation
  - Resolution

Image Representation

- Before we discuss image acquisition recall that a digital image is composed of M rows and N columns of pixels each storing a value
- Pixel values are most often grey levels in the range 0-255(black-white)
- We will see later on that images can easily be represented as matrices



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Sensing

- Incoming energy lands on a sensor material responsive to that type of energy and this generates a voltage
- Collections of sensors are arranged to capture images

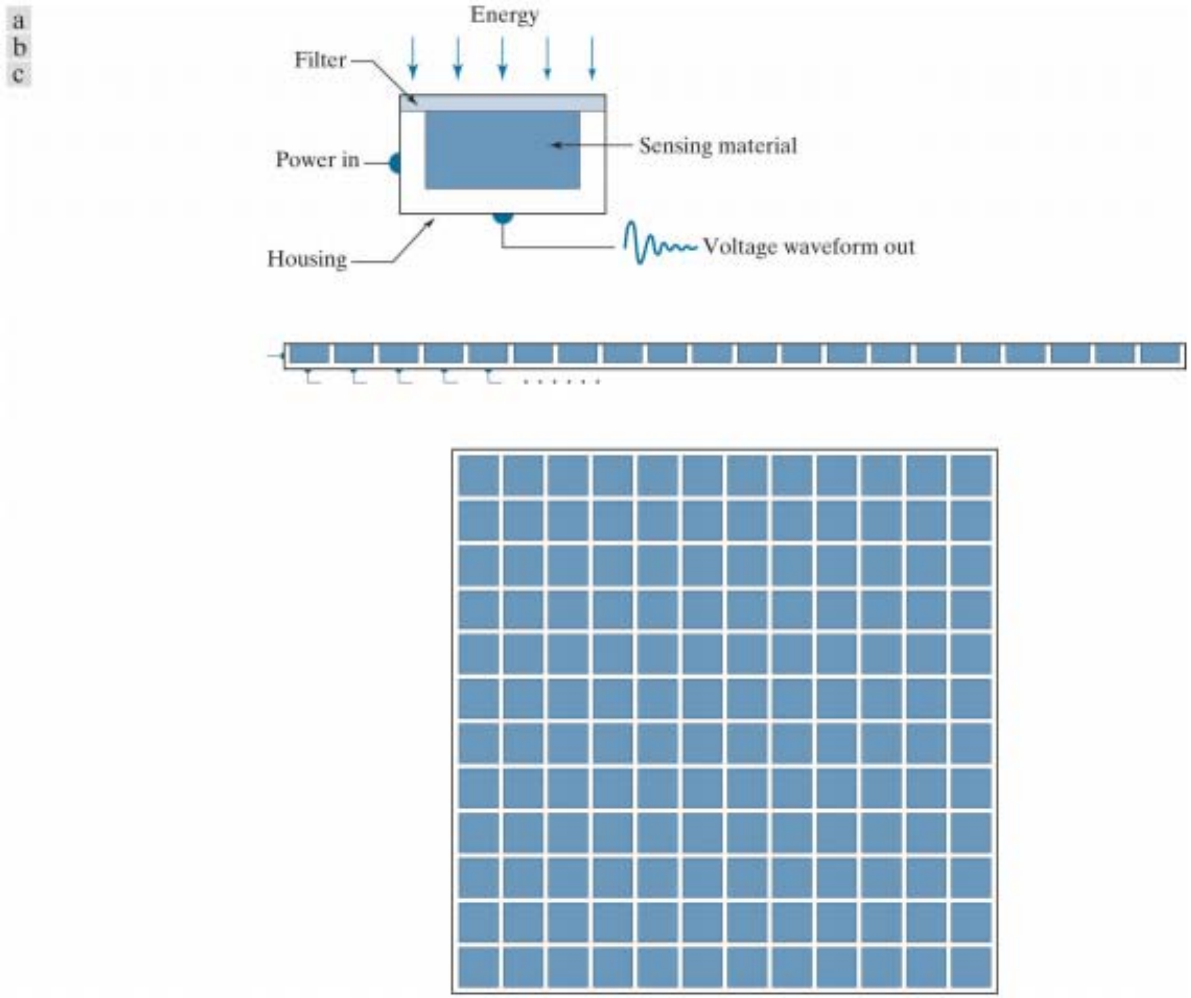


FIGURE 2.12
(a) Single sensing element. (b) Line sensor. (c) Array sensor.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Acquisition Using a Single Sensing Element

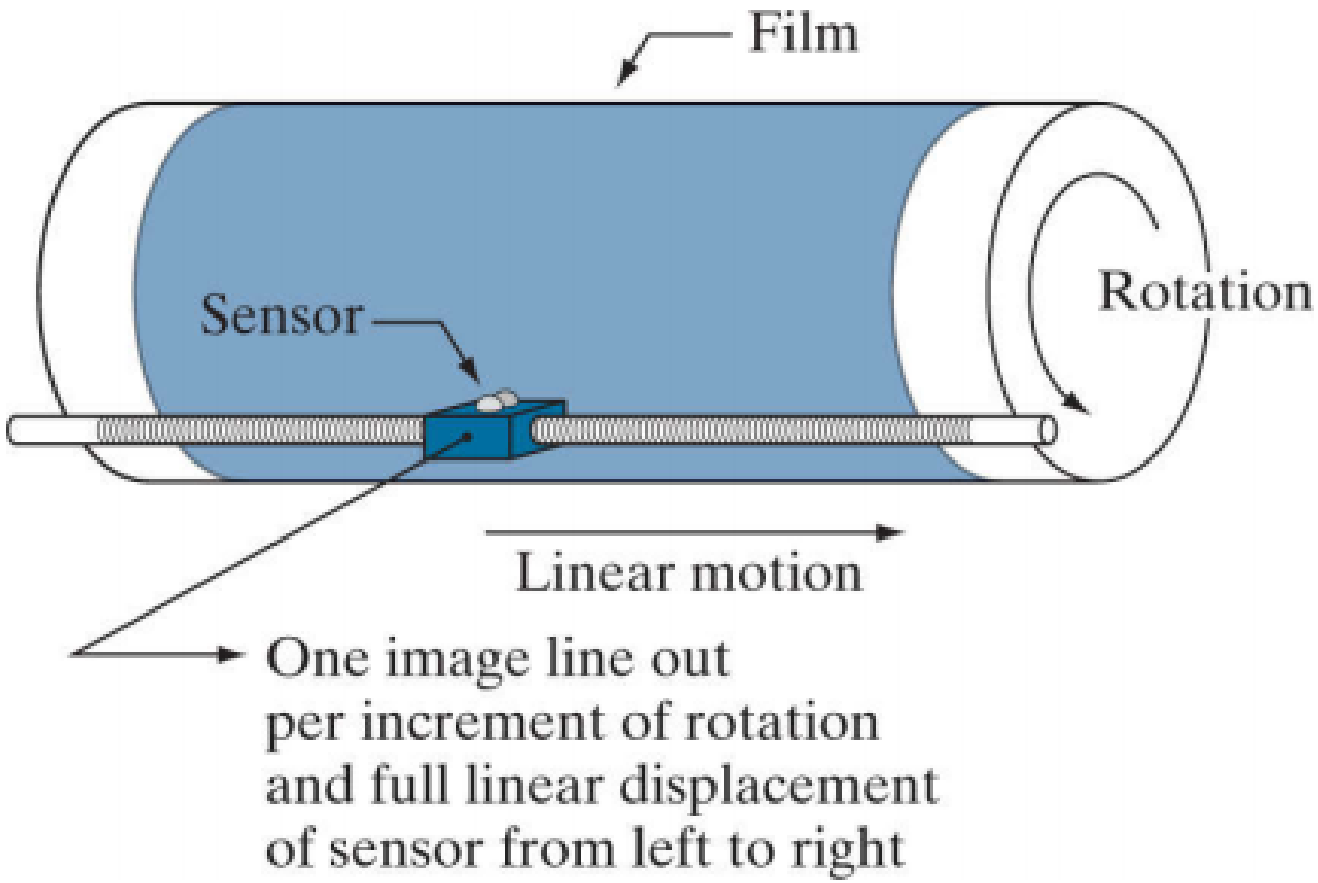


FIGURE 2.13
Combining a single sensing element with mechanical motion to generate a 2-D image.

Image Sensing

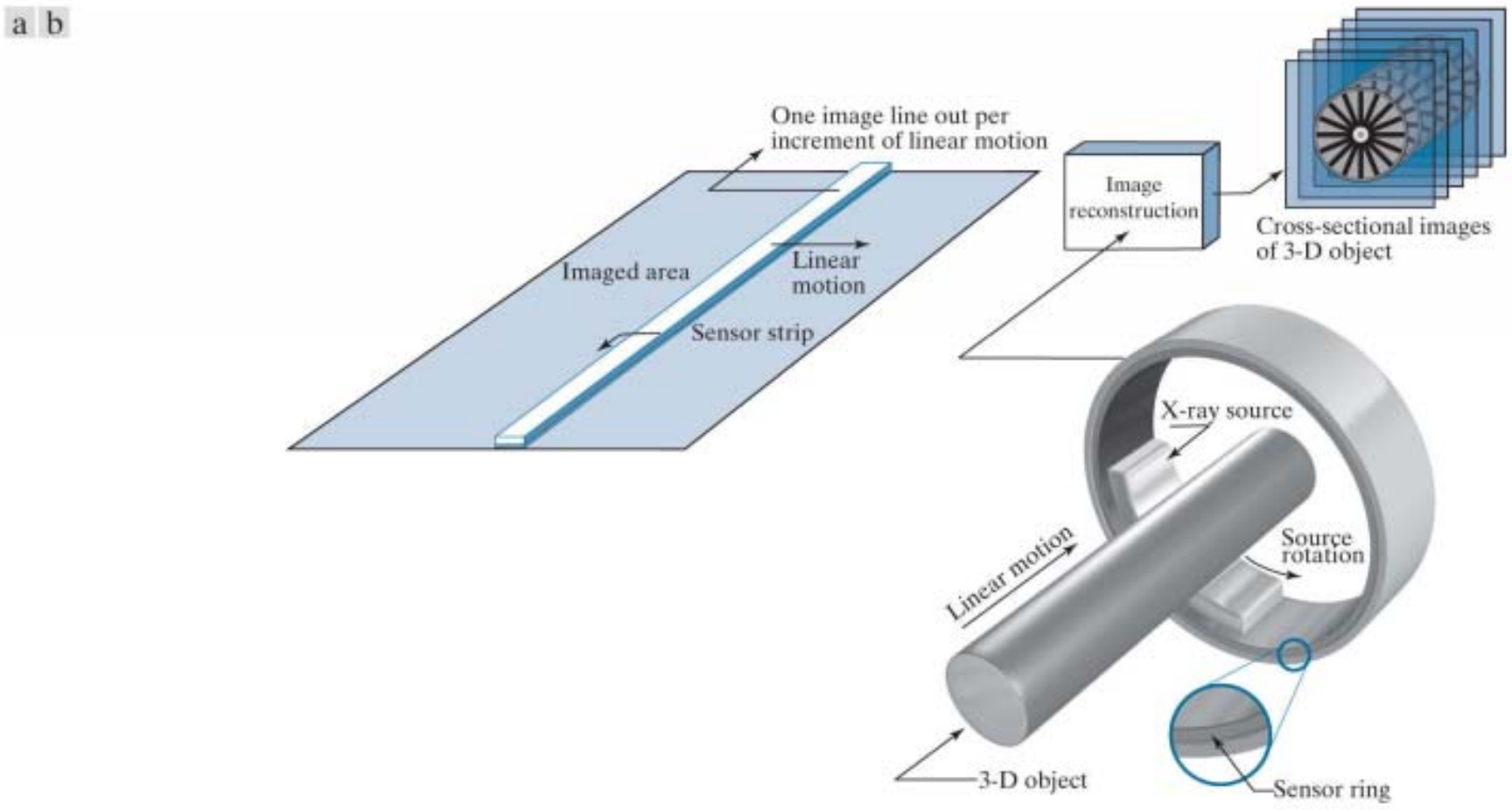
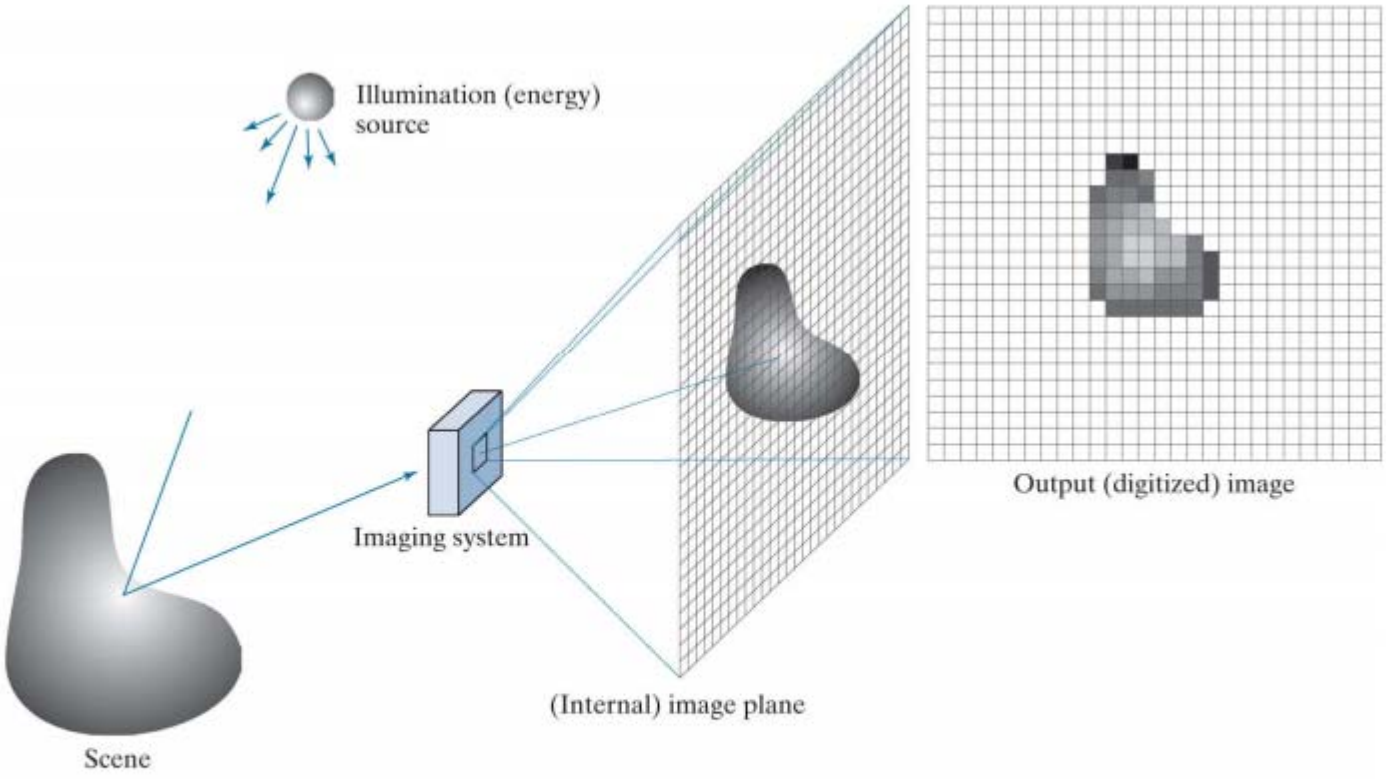


FIGURE 2.14
(a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Acquisition

- Images are typically generated by *illuminating* a *scene* and absorbing the energy reflected by the objects in that scene



a b c d e

FIGURE 2.15
An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

A Simple Image Formation Model

$$0 \leq f(x,y) < \infty$$

- Function f(x, y) is characterized by two components:
  1. the amount of source illumination incident on the scene being viewed, and
  2. the amount of illumination reflected by the objects in the scene. Appropriately, these are called the illumination and reflectance components, and are denoted by i(x, y) and r(x, y), respectively.

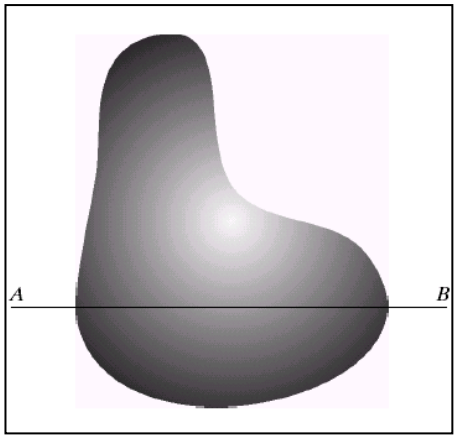
  The two functions combine as a product to form f(x, y):

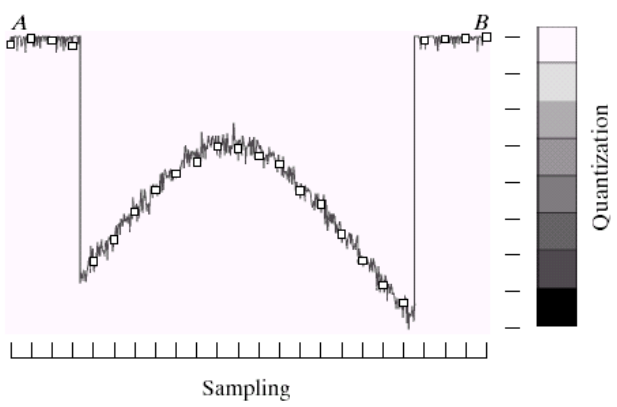
$$f(x,y) = i(x,y)r(x,y)$$

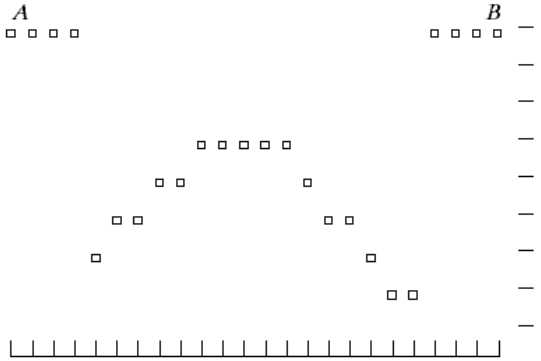
$$0 \leq i(x,y) < \infty \qquad 0 \leq r(x,y) \leq 1$$

Image Sampling And Quantisation

- A digital sensor can only measure a limited number of **samples** at a **discrete** set of energy levels
- *Quantisation* is the process of converting a continuous **analogue** signal into a digital representation of this signal

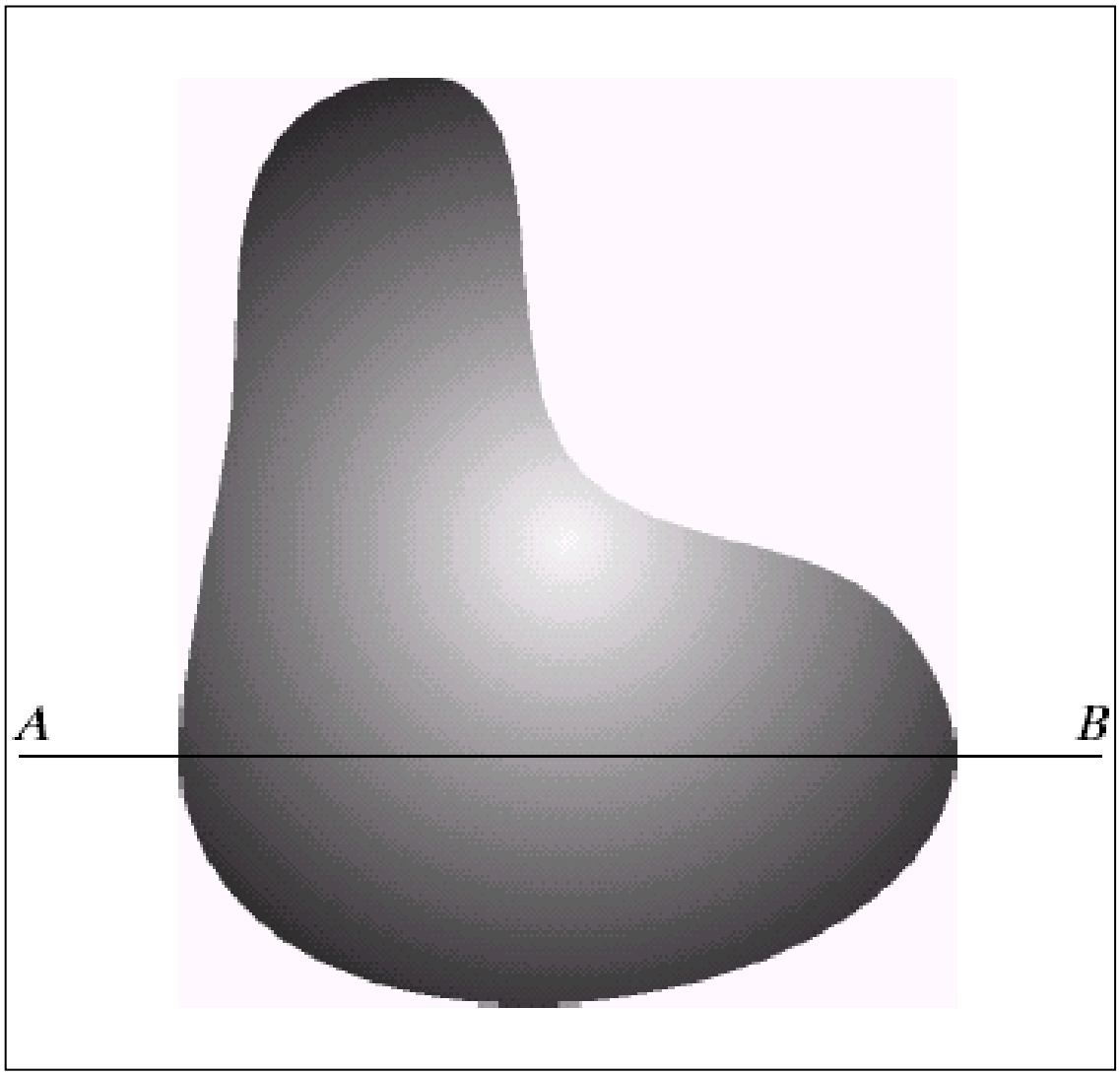






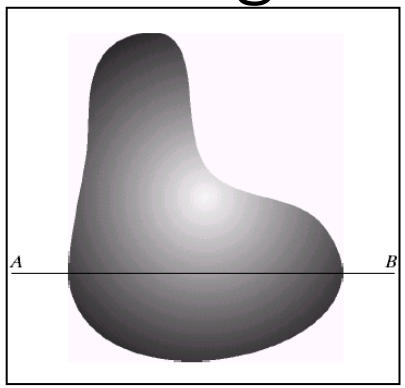
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

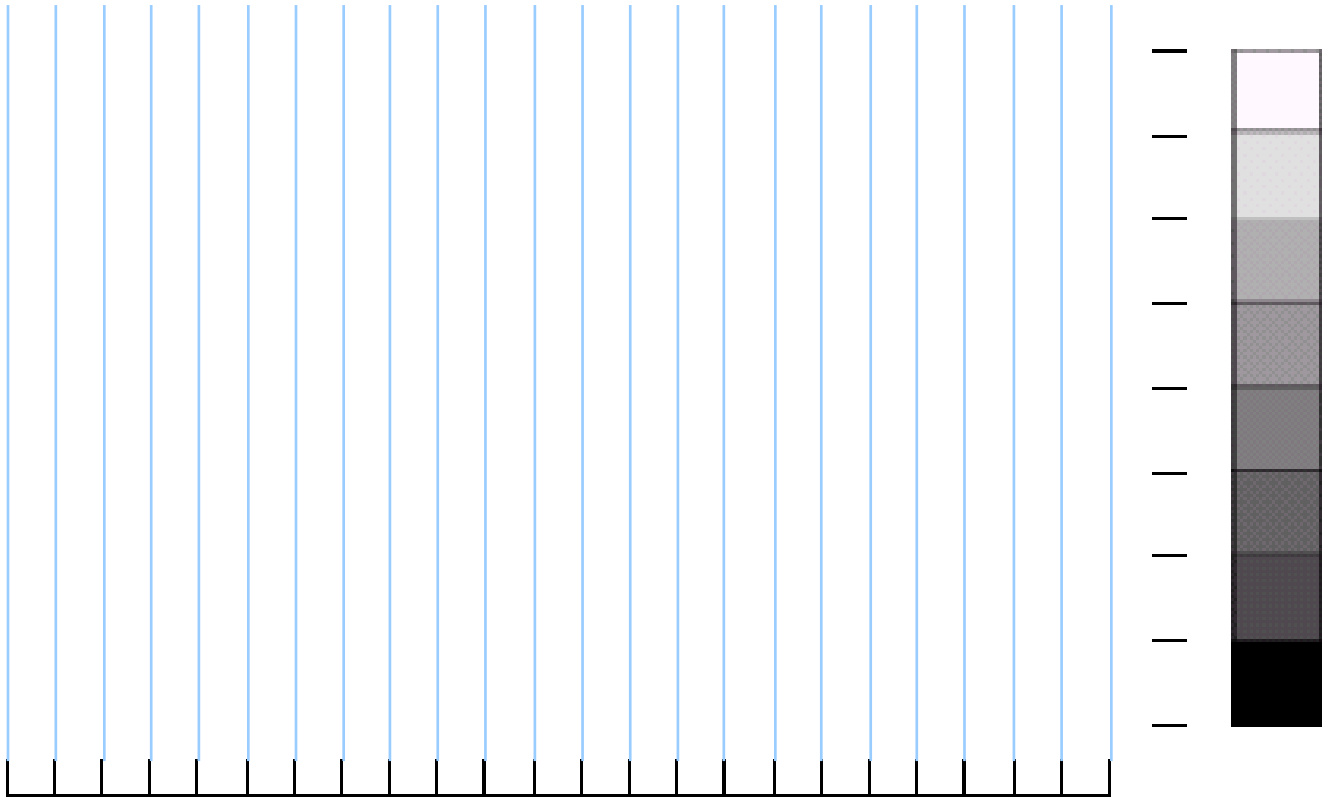
Image Sampling And Quantisation



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Sampling And Quantisation





Sampling

Image Sampling And Quantisation (cont…)

•Remember that a digital image is always only an **approximation** of a real world scene

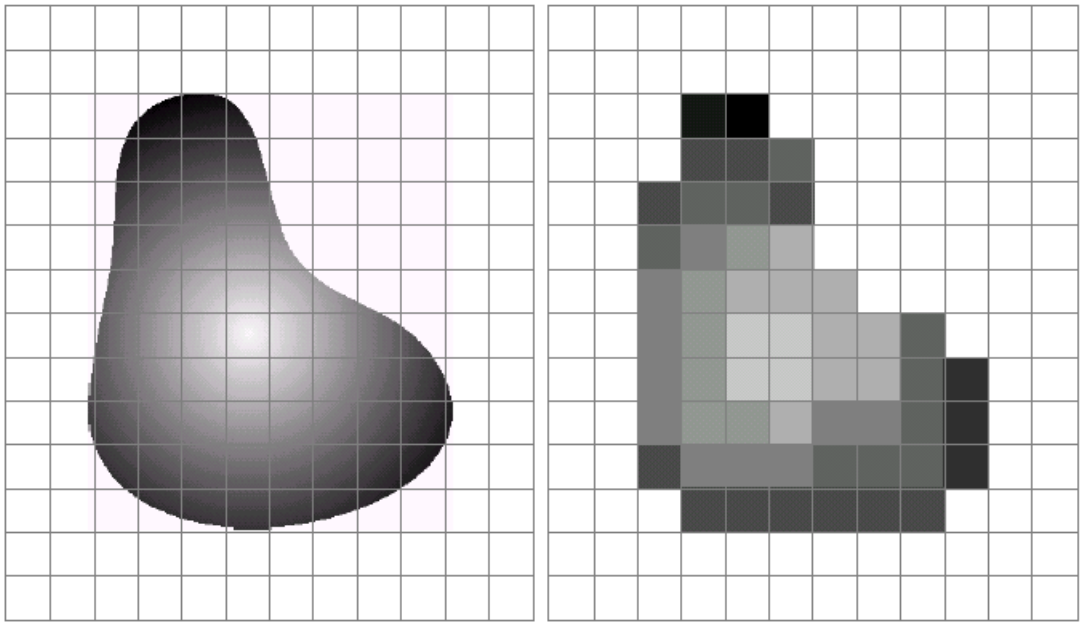
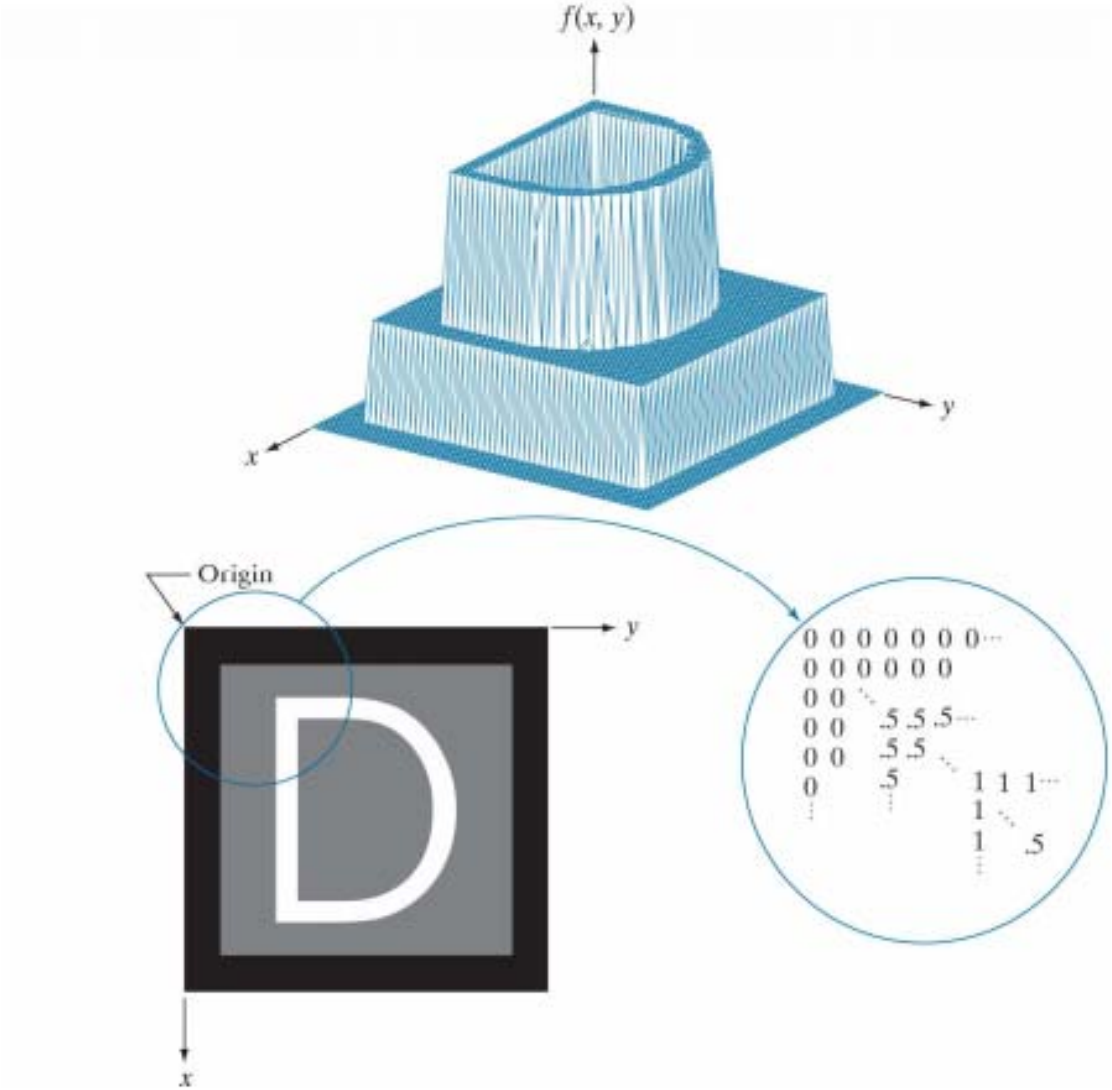




Image Representation



Representing Digital Images

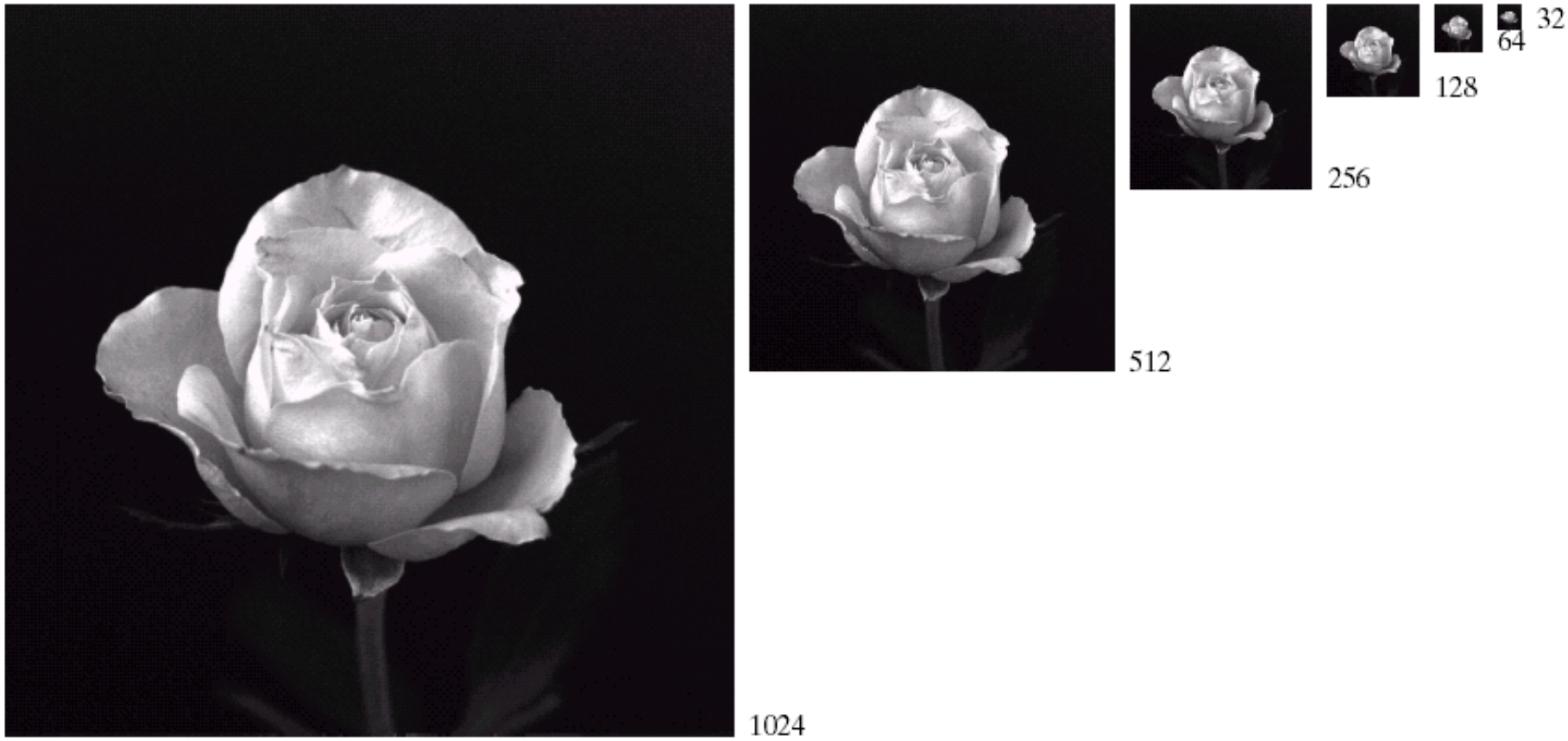
$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

Spatial Resolution

- *The spatial resolution* of an image is determined by how sampling was carried out
- Spatial resolution simply refers to the smallest discernable detail in an image
  - Vision specialists will often talk about pixel size
  - Graphic designers will talk about *dots per inch* (DPI)



Spatial Resolution (cont…)



Spatial Resolution (cont…)



Spatial Resolution (cont…)



Spatial Resolution (cont…)



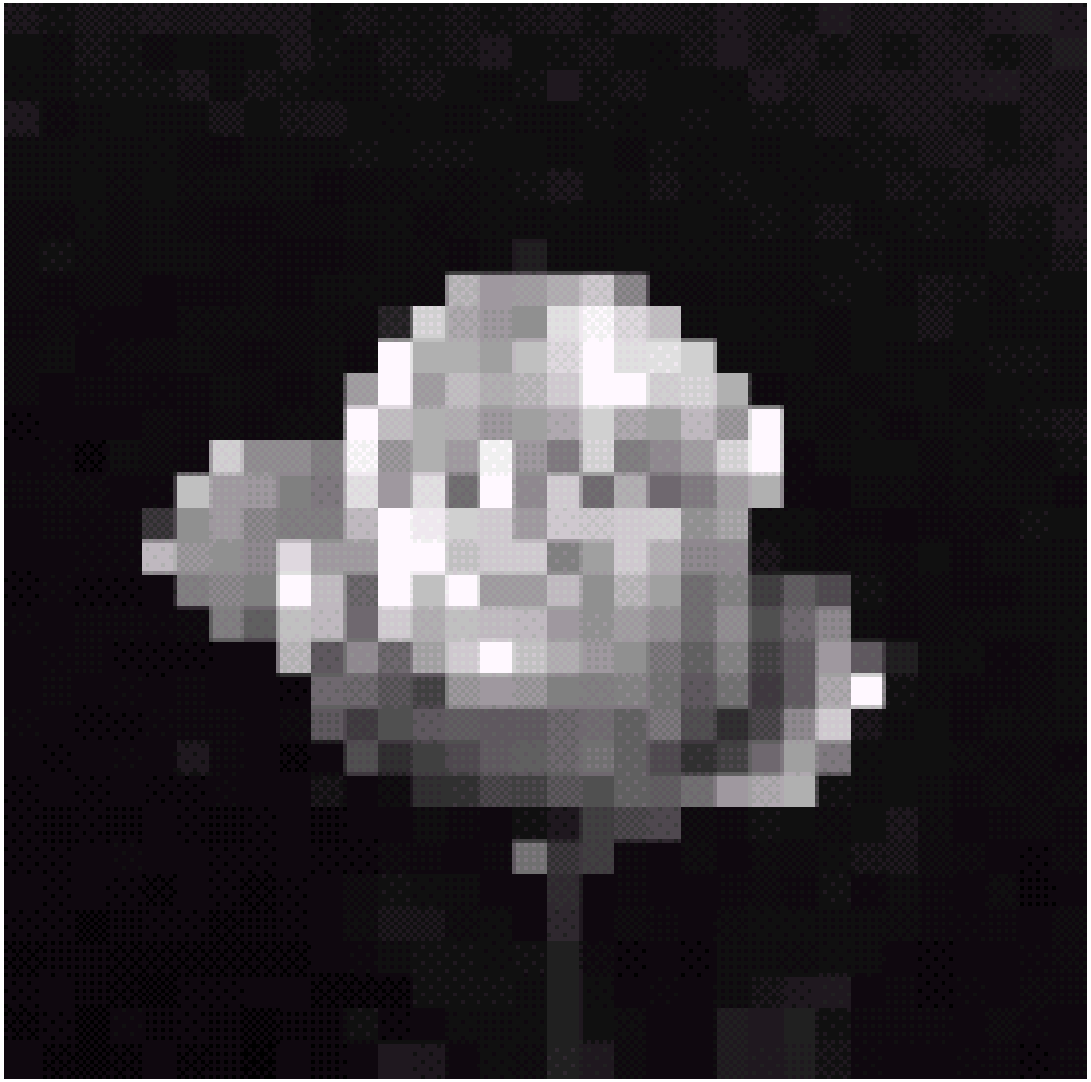
Spatial Resolution (cont…)



Spatial Resolution (cont…)



Spatial Resolution (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

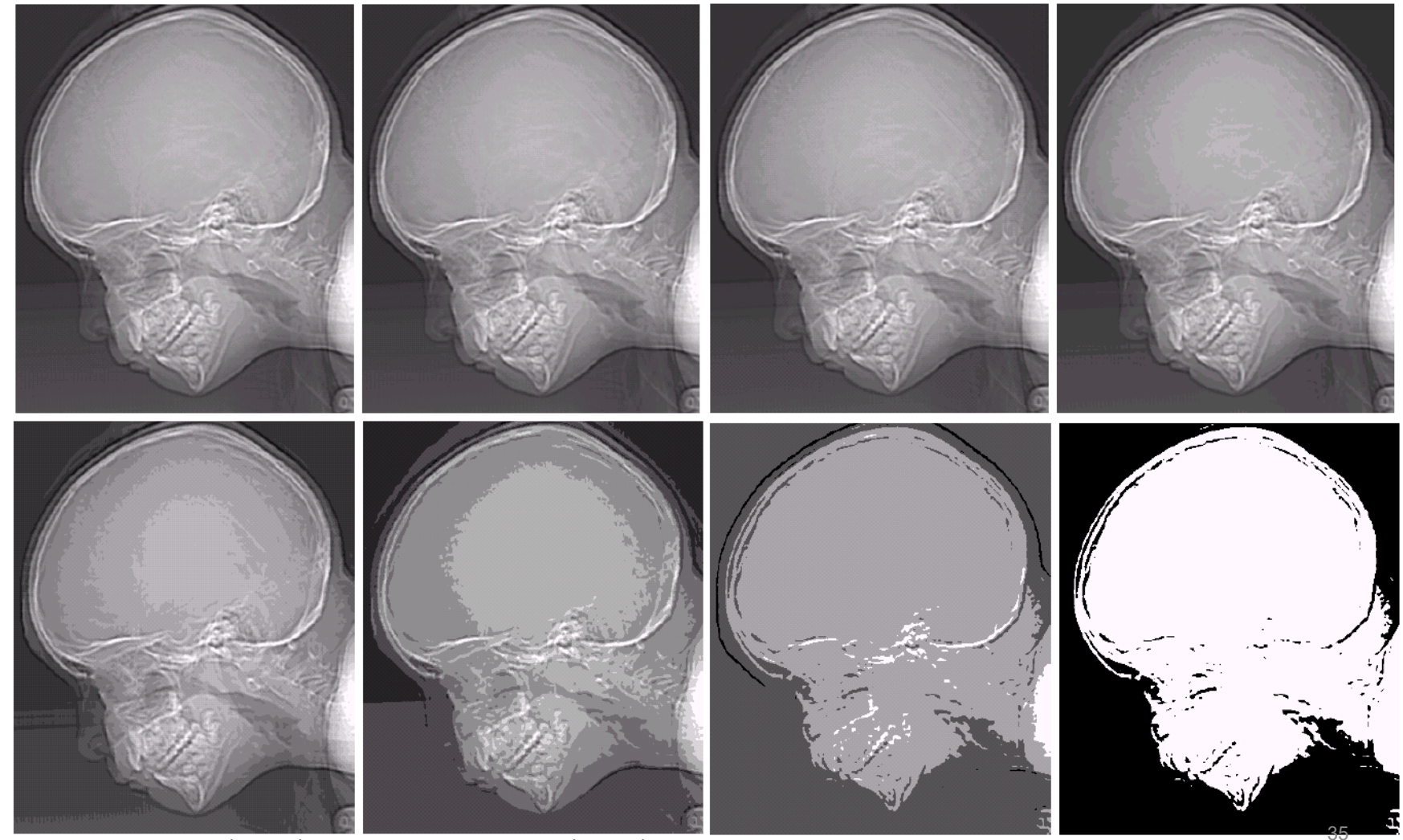
Intensity Level Resolution

•*Intensity level resolution* refers to the number of intensity levels used to represent the image

- The more intensity levels used, the finer the level of detail discernable in an image
- Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



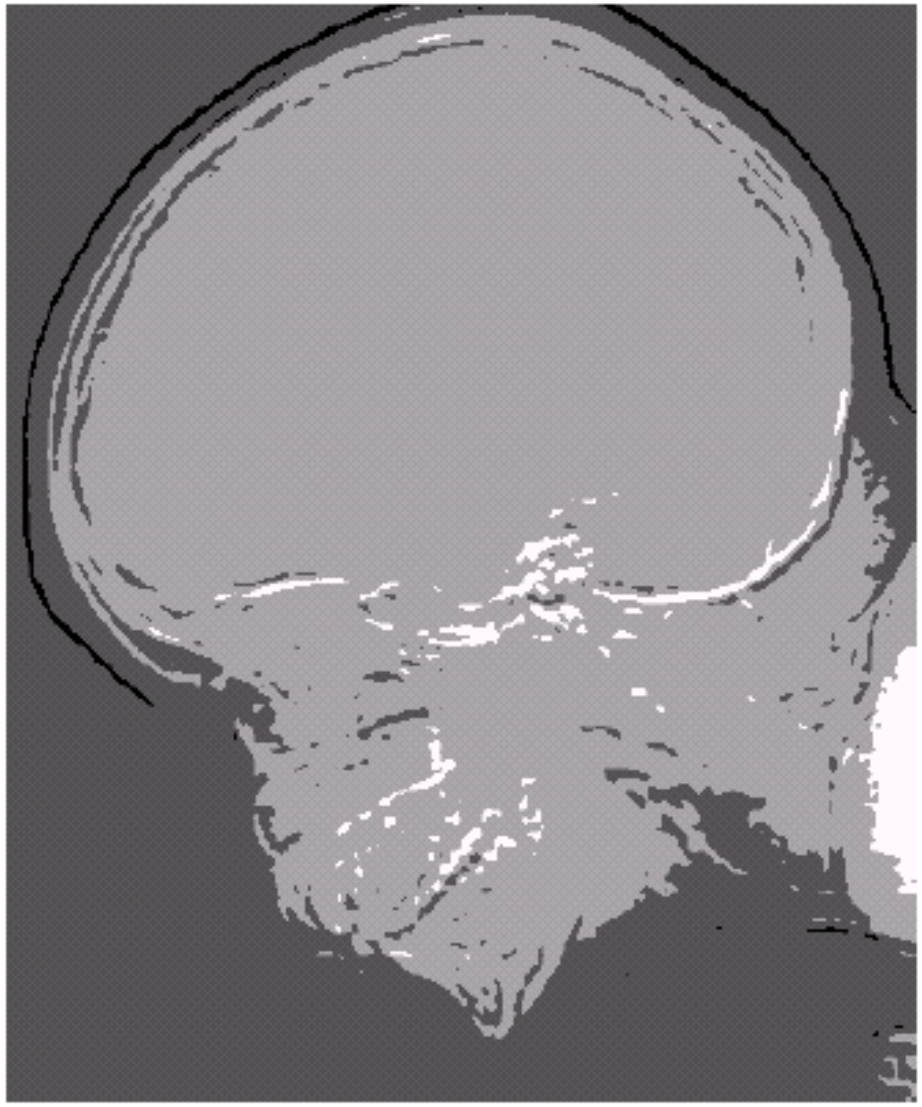
Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Saturation & Noise

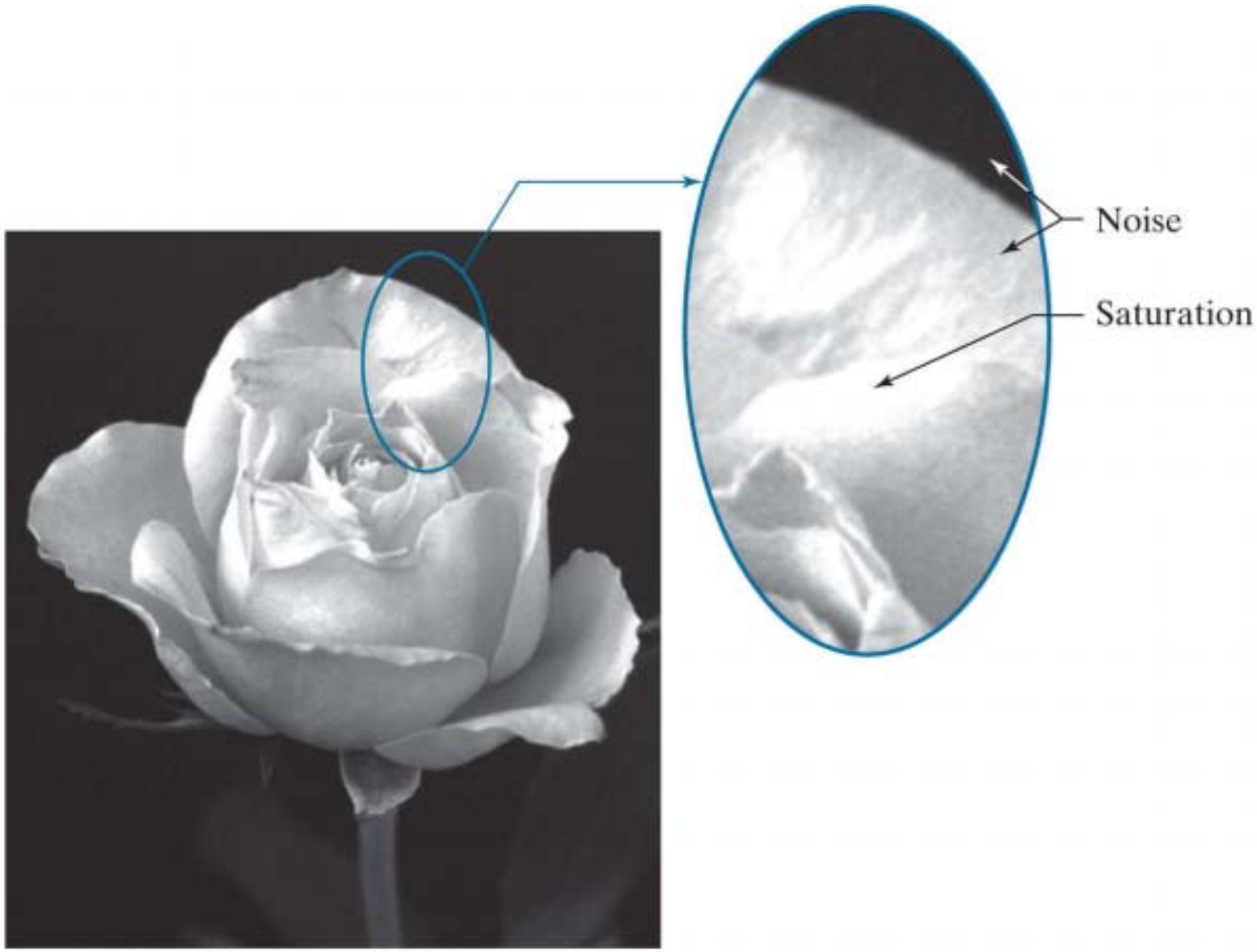


FIGURE 2.20
An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity values are clipped (note how the entire saturated area has a high, constant intensity level). Visible noise in this case appears as a grainy texture pattern. The dark background is noisier, but the noise is difficult to see.

Spatial and Intensity Resolution

- spatial resolution is a measure of the smallest discernible detail in an image
- Quantitatively, spatial resolution can be stated in several ways, with line pairs per unit distance, and dots (pixels) per unit distance

a b
c d



FIGURE 2.23

Effects of reducing spatial resolution. The images shown are at: (a) 930 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi.

Resolution: How Much Is Enough?

- The big question with resolution is always *how much is enough*?
- This all depends on what is in the image and what you would like to do with it

  Key questions include
  - Does the image look aesthetically pleasing?
  - Can you see what you need to see within the image?

Resolution: How Much Is Enough? (cont…)





•The picture on the right is fine for counting the number of cars, but not for reading the number plate

Intensity Level Resolution (cont…)



Low Detail



Medium Detail



High Detail

Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Intensity Level Resolution (cont…)



Storage Requirements

- Consider an NxN image having k bits per pixel.
- Color (RGB) images require three times the storage (assuming no compression).

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Image Interpolation

- image resizing (shrinking and zooming), which are basically image resampling
- Interpolation is the process of using known data to estimate values at unknown locations.

Some Basic Relationships Between Pixels

A pixel *p* at coordinates (*x*, *y*) has two horizontal and two vertical neighbors with coordinates

$$(x+1, y), (x-1, y), (x, y+1), (x, y-1)$$

This set of pixels, called the 4-*neighbors* of *p*, is denoted $N_4(p)$.

The four *diagonal* neighbors of *p* have coordinates

$$(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$$

and are denoted $N_D(p)$These neighbors, together with the 4-neighbors, are called the 8-neighbors of p, denoted by $N_8(p)$The set of image locations of the neighbors of a point p is called the neighborhood of p

Adjacency, Connectivity, Regions, and Boundaries

- Let V be the set of intensity values used to define adjacency. In a binary image, if we are referring to adjacency of pixels with value 1.
- In a grayscale image, set V typically contains more elements. For example, if we are dealing with the adjacency of pixels whose values are in the range 0 to 255, set V could be any subset of these 256 values. We consider three types :
- *4-adjacency:* Two pixels p and q with the values from set 'V' are 4-adjacent if q is in the set of $N_4(p)$.
- *8-adjacency:* Two pixels p and q with the values from set 'V' are 8-adjacent if q is in the set of $N_8(p)$.

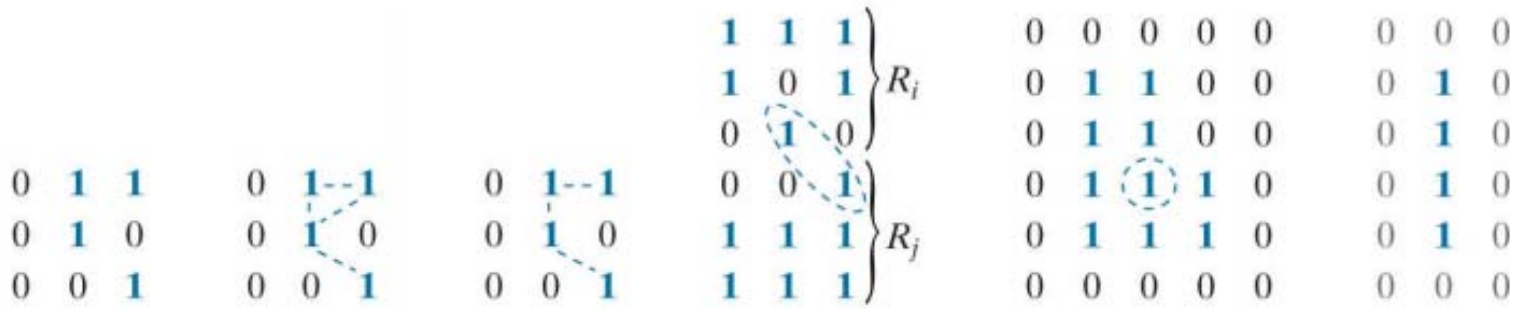
Adjacency, Connectivity, Regions, and Boundaries

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

(i) q is in $N_4(p)$ OR

(ii) q is in $N_D(p)$ & the set $N_4(p) \cap N_4(q)$ have no pixels whose values are from 'V'.

e.g. V = { 1 }



a b c d e f

FIGURE 2.28

(a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

Distance Measures

- For pixels p, q, and s, with coordinates (x, y), (u, v), and (w, z), respectively, D is a distance function or metric if

a. $D(p,q) \geq 0 \quad (D(p,q) = 0 \quad \text{iff} \quad p = q)$,

b. $D(p,q) = D(q,p)$, and

c. $D(p,s) \leq D(p,q) + D(q,s)$.

Distance Measures

- The Euclidean distance between p and q is defined as

$$D_e(p,q) = [(x-u)^2 + (y-v)^2]^{\frac{1}{2}}$$

Distance Measures

City Block Distance: The D_4 distance between p & q is defined as

$$D_4(p, q) = |x - s| + |y - t|$$

In this case, pixels having D_4 distance from (x, y) less than or equal to some value r form a diamond centered at (x, y).

```
        2
    2   1   2
2   1   0   1   2
    2   1   2
        2
```

Pixels with D_4 distance ≤ 2 forms the following contour of constant distance.

Distance Measures

Chess-Board Distance: The D_8 distance between p & q is defined as

$$D_8(p, q) = \max(\ |x - s|\ ,\ |y - t|\ )$$

In this case, pixels having D_8 distance from (x, y) less than or equal to some value r form a square centered at (x, y).

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Pixels with D_8 distance ≤ 2 forms the following contour of constant distance.

Summary

- We have looked at:
  - Human visual system
  - Light and the electromagnetic spectrum
  - Image representation
  - Image sensing and acquisition
  - Sampling, quantisation and resolution
- Next time we start to look at techniques for image enhancement